Bayesian Inference

INFO 521 · Module 4 · Lecture B — The Conjugate Gaussian Posterior

Greg Chism

School of Information · University of Arizona

July 2026

Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of Lecture B you should be able to:

1. **Place a Gaussian prior on the weights** and state the conjugate posterior $\mathcal{N}(\mathbf{w} \mid \mathbf{m}_N, \mathbf{S}_N)$.
2. **Read the update formulas** — what shrinks, what dominates, and why $\mathbf{X}^\top\mathbf{X}$ appears *again*.
3. **Show ridge regression is MAP** with a Gaussian prior ($\lambda = \sigma^2/\tau^2$).
4. **Form the posterior predictive** and separate its two variance terms.

From θ to w

Lecture A put a distribution on a *proportion*. Now put one on the **regression weights** of the Module-1 model:

$$\text{prior:}\quad p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_0,\ \mathbf{S}_0) \qquad\qquad \text{likelihood:}\quad p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^{N} \mathcal{N}\big(y_n \,\big|\, \mathbf{w}^\top \mathbf{x}_n,\ \sigma^2\big)$$

- $\mathbf{m}_0$ — where belief starts (e.g., “SBP near 120, no age effect”).
- $\mathbf{S}_0$ — how *loosely* it starts (wide diagonal = weakly informative).
- σ^2 — treated as **known** today (use m3a’s $\hat{\sigma}^2$); it keeps the algebra conjugate.

The conjugate Gaussian posterior

Gaussian prior × Gaussian likelihood → **Gaussian posterior**, in closed form:

$$p(\mathbf{w} \mid \mathcal{D}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_N,\ \mathbf{S}_N), \qquad \boxed{\mathbf{S}_N = \left(\mathbf{S}_0^{-1} + \tfrac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}\right)^{-1}, \quad \mathbf{m}_N = \mathbf{S}_N \left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \tfrac{1}{\sigma^2}\mathbf{X}^\top\mathbf{y}\right)}$$

Project 1 ends exactly here. The full pipeline — data → model → loss → likelihood → posterior — closes on this pair of formulas.

Reading the formulas

Work in **precisions** (inverse covariances) and the update is addition again:

- $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}$ — **precisions add**: prior information + data information.
- The data's contribution $\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}$ is the **Fisher information** $\mathcal{I}$ from m3b — the same object, fourth appearance.
- $\mathbf{m}_N$ is a **precision-weighted blend** of $\mathbf{m}_0$ and the data — the vector version of m4a's shrinkage identity.
- $N \to \infty$: the data term dominates, $\mathbf{m}_N \to \hat{\mathbf{w}}_{\mathrm{MLE}},\ \mathbf{S}_N \to \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \mathrm{Cov}[\hat{\mathbf{w}}]$.

Watching the posterior contract

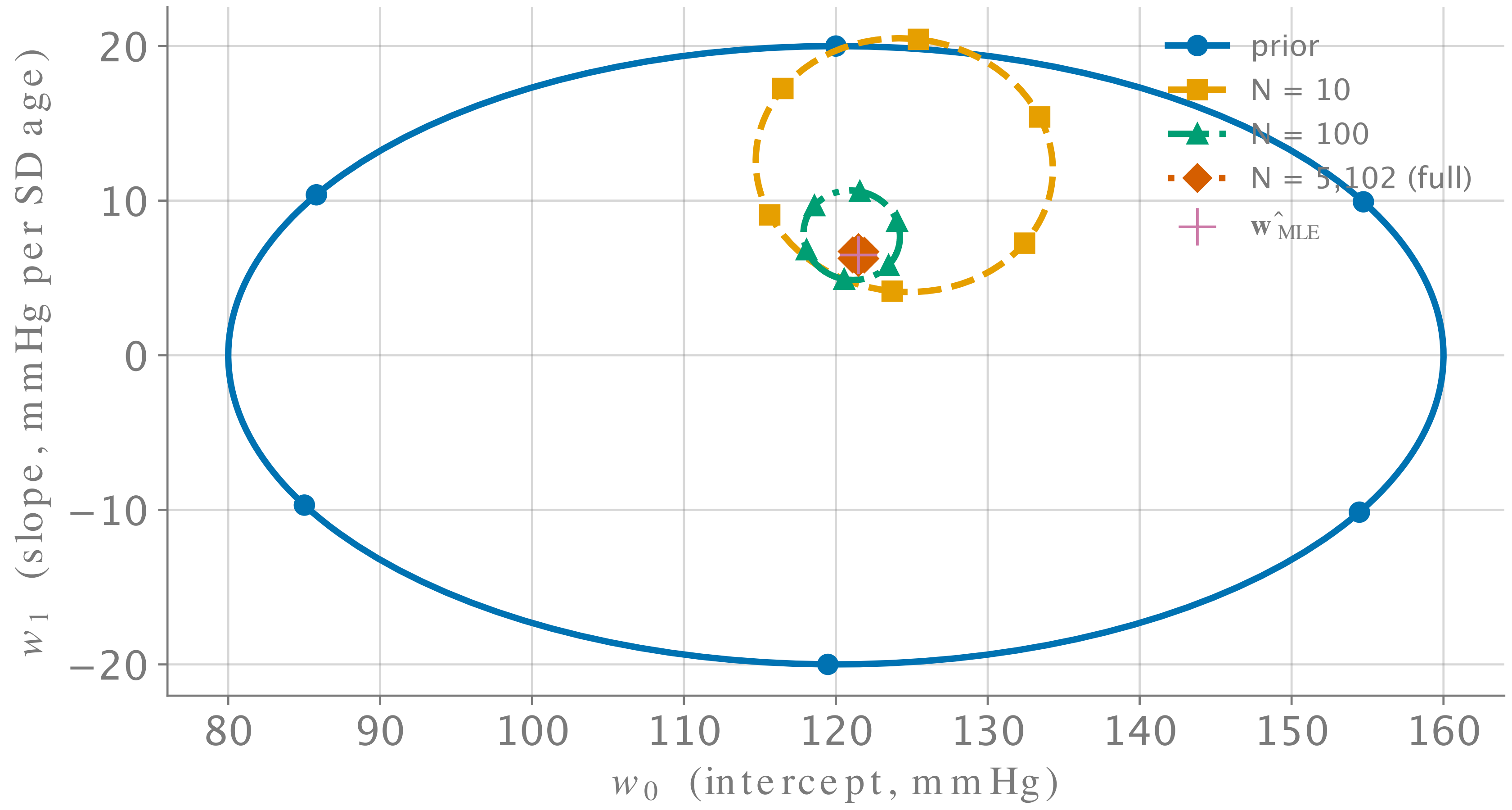

Figure 1: Conjugate posterior over (w_0, w_1), real NHANES draws (seed 521): prior → N = 10 → 100 → 5,102. The purple + is the MLE.

Special case: the ridge prior

Take a **zero-mean isotropic** prior, $\mathbf{m}_0 = \mathbf{0}$, $\mathbf{S}_0 = \tau^2 \mathbf{I}$:

$$\mathbf{m}_N = \left(\mathbf{X}^\top \mathbf{X} + \tfrac{\sigma^2}{\tau^2}\mathbf{I}\right)^{-1} \mathbf{X}^\top \mathbf{y} \;=\; \hat{\mathbf{w}}_{\text{ridge}} \quad \text{with} \quad \boxed{\lambda = \frac{\sigma^2}{\tau^2}}$$

- m2b's forward hook, **cashed in**: the ridge penalty *is* a Gaussian prior.
- λ stops being a dial and becomes a *statement*: noisy data (big σ^2) or a tight prior (small τ^2) ⇒ shrink harder.

MAP vs. the full posterior

The **MAP** (maximum a posteriori) estimate is the posterior's peak:

$$\hat{\mathbf{w}}_{\mathrm{MAP}} = \arg\max_{\mathbf{w}} \big[\ell(\mathbf{w}) + \log p(\mathbf{w})\big] \;=\; \mathbf{m}_N \quad (\text{Gaussian case: peak} = \text{mean}).$$

- **MAP is one number** — regularized fitting in Bayesian clothes; it discards the spread.
- **The posterior is the answer** — $\mathbf{S}_N$ carries what MAP throws away, and predictions (next slide) *use* it.
- Gaussians hide the difference (peak = mean); Module 5's posteriors won't be Gaussian, and the gap becomes the whole story.

Predicting a new patient

For a new input $\mathbf{x}_*$, integrate the model over the posterior:

$$p(y_* \mid \mathbf{x}_*, \mathcal{D}) = \mathcal{N}\left(y_* \,\middle|\, \mathbf{m}_N^\top \mathbf{x}_*,\ \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\mathbf{x}_*^\top \mathbf{S}_N \, \mathbf{x}_*}_{\text{parameter uncertainty}} \right)$$

Two variance terms, two different fixes:

- σ^2 — patient-level scatter; **irreducible** (m2b's floor).
- $\mathbf{x}_*^\top \mathbf{S}_N \mathbf{x}_*$ — *our* ignorance of $\mathbf{w}$; **shrinks with data**, grows away from the data's center.

The predictive band on NHANES

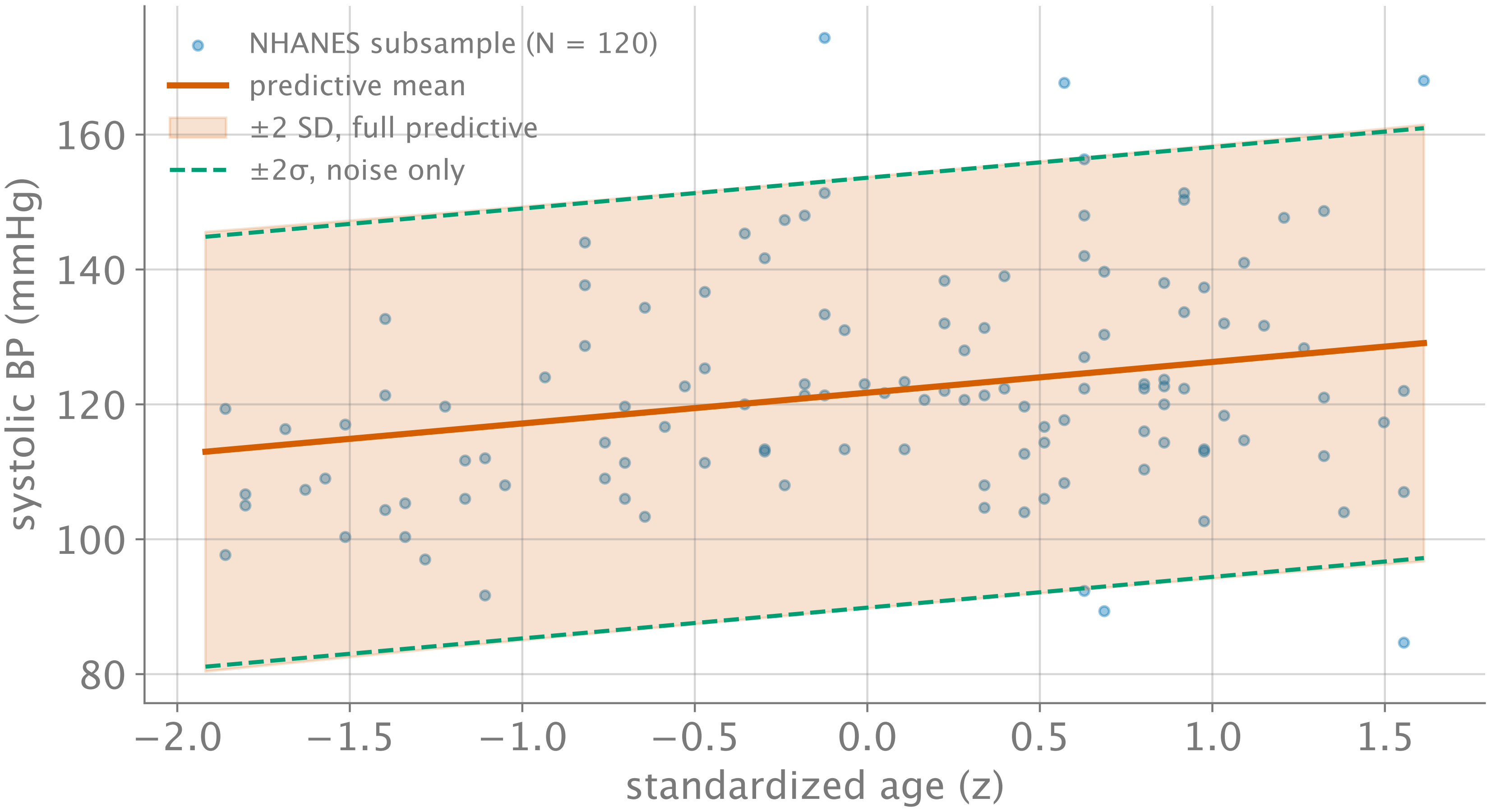

Figure 2: Posterior predictive from N = 120 real patients: mean ±2 predictive SD (shaded) vs. noise-only ±2σ (green dashes) — parameter uncertainty flares at the extremes.

The evidence, briefly

m4a's ignored denominator, $p(\mathcal{D})$ — the **marginal likelihood** or *evidence* — is the likelihood **averaged over the prior**:

$$p(\mathcal{D}) = \int p(\mathcal{D} \mid \mathbf{w})\, p(\mathbf{w})\, d\mathbf{w}$$

- Scores a **whole model** (prior included), not one parameter value.
- Comparing evidences ranks models **without a test set** — an automatic Occam's razor: too-flexible models spread their prior thin and score lower.
- For today's conjugate model it, too, is closed-form; beyond, it is the first casualty (Module 5).

Recap & bridge

- **Conjugate Gaussian posterior**: $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X},\ \mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{y})$ — **Project 1's endpoint**.
- **Ridge = MAP** with $\mathbf{S}_0 = \tau^2\mathbf{I}$: $\lambda = \sigma^2/\tau^2$.
- **Predictive variance** $= \sigma^2 + \mathbf{x}_*^\top\mathbf{S}_N\mathbf{x}_*$ — noise + ignorance, separately visible.

Next (Module 5): binarize the outcome — hypertensive or not — and the Gaussian likelihood is gone. Conjugacy **breaks**, no closed form survives, and approximation becomes the craft.